Beyond Dopamine: Shared Neurobiology of Behavioral Addictions

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Abstract

Behavioral addictions are neuropsychiatric disorders characterized by the uncontrollable and repetitive engagement in specific behaviors without the use of any psychoactive substance. In recent years, increasing interest in behavioral addictions such as gambling, gaming, social media, food, shopping, romantic love- attachment, and compulsive sexual behavior disorder has enabled a more comprehensive investigation of their common neurobiological mechanisms. Current evidence indicates that behavioral addictions cannot be explained solely by the dopaminergic reward system; glutamatergic, GABAergic, serotonergic, endocannabinoid, and endogenous opioid systems, as well as oxytocin, vasopressin, and stress-related mechanisms, also play important roles in the development of addiction. In addition, alterations in the reward circuitry, frontostriatal networks, synaptic plasticity, and executive control mechanisms constitute the fundamental neurobiological basis underlying the emergence and persistence of compulsive behaviors.

In this narrative review, the neurobiological foundations of behavioral addictions are evaluated in light of the current literature, with particular emphasis on the roles of the reward system and major neurotransmitter networks in the development of addiction. Furthermore, the principal types of behavioral addictions, including gambling disorder, gaming disorder, social media addiction, compulsive sexual behavior disorder, romantic love- attachment, food addiction, and shopping addiction, are comparatively examined, and their shared as well as distinct neurobiological characteristics are discussed. Finally, emerging areas of research, including neuroinflammation, the gut–brain axis, glucagon-like peptide-1 receptor agonists, the gut microbiota, and artificial intelligence-driven digital environments, are reviewed with respect to their influence on addiction. Potential future developments in biomarker-based diagnostic approaches and personalized treatment strategies are also discussed. A better understanding of the multidimensional neurobiological mechanisms underlying behavioral addictions may contribute substantially to improving diagnostic accuracy and to the development of novel targeted therapeutic strategies.

Keywords: behavioral addiction, neurobiology, reward system, dopamine, glutamate, GABA, serotonin, endocannabinoid system, neuroplasticity, behavioral addictions

1. Introduction

Addiction is regarded as a chronic brain disorder with biological, psychological, and social dimensions, characterized by the individual's persistent and uncontrollable engagement in the use of a particular substance or repetition of a particular behavior despite being aware of its harmful consequences. In the past, the concept of addiction was largely focused on the use of alcohol, nicotine, and other psychoactive substances; however, neurobiological and clinical evidence obtained in recent years has demonstrated that certain behaviors occurring in the absence of substance use may also acquire addictive characteristics as a result of similar neuroadaptive processes. In particular, the classification of gambling disorder as a behavioral addiction in the DSM-5 and ICD-11 has demonstrated that the concept of addiction is not limited solely to chemical substances and has contributed to an acceleration of research into the neurobiological bases of other behavioral addictions, such as gaming addiction, problematic social media use, compulsive sexual behavior disorder, food addiction, and compulsive buying behavior. [28,31,35,49]

Structural and functional changes occurring in brain networks responsible for reward processing, motivation, learning, impulse control, and decision-making are thought to play a central role in the development of behavioral addictions. Interactions among the ventral tegmental area (VTA), nucleus accumbens (NAc), prefrontal cortex (PFC), amygdala, and hippocampus, particularly within the mesolimbic dopaminergic system, constitute fundamental mechanisms underlying the formation of reward anticipation, craving, reinforcement of behavior, and the development of relapse. Nevertheless, current research has demonstrated that the neurobiology of addiction cannot be explained solely by the dopaminergic system; in addition to the glutamatergic, GABAergic, serotonergic, endocannabinoid, and endogenous opioid systems, oxytocin, vasopressin, and stress-related neuropeptides also interact with one another within this complex neural network. Therefore, addiction is considered a dynamic process involving the combined contribution of numerous neurochemical and neuroanatomical mechanisms rather than a dysfunction of a single neurotransmitter system. [2,4,9,13,16,19,22,24]

The increasing prevalence of behavioral addictions in society has reached an even more prominent dimension as digital technologies have become an integral part of everyday life. Online gaming platforms, social media applications, digital gambling systems, and personalized content algorithms may constitute environmental factors that continuously stimulate the reward circuitry, thereby increasing the risk of developing addiction, particularly among children and young adults. In addition, behavioral addictions frequently co-occur with depression, anxiety disorders, attention-deficit/hyperactivity disorder, obsessive-compulsive symptoms, and other psychiatric disorders, and may significantly adversely affect an individual’s academic achievement, occupational performance, social relationships, and quality of life. Nevertheless, efforts to standardize diagnostic criteria, identify biomarkers, and develop effective treatment options for many types of behavioral addictions are still ongoing. [31,35,49–51]

In recent years, addiction research has moved beyond the classical reward system approach to also encompass novel biological mechanisms such as neuroinflammation, the gut-brain axis, metabolic signaling, and glucagon-like peptide-1 (GLP-1) receptor agonists. At the same time, understanding the effects of artificial intelligence-supported digital platforms on user behavior has made the importance of environmental and technological factors in the development of addiction even more apparent. These developments support the consideration of addiction as a multidimensional neurobiological disorder, while also creating new areas of research aimed at developing biomarker-based diagnostic methods and personalized pharmacological and behavioral treatment strategies. [52–56]

The aim of this review is to comprehensively evaluate the current neurobiological mechanisms of behavioral addictions; to explain the roles of the reward system and major neurotransmitter networks in the development of addiction; to comparatively examine the common and distinct neurobiological characteristics of the major types of behavioral addictions; and to present the novel biological approaches that have emerged in recent years, as well as future-oriented diagnostic and treatment strategies, in accordance with the current literature.

2. Methods

This review was prepared to evaluate the current evidence regarding the neurobiological foundations of behavioral addictions. The literature search was conducted exclusively in the PubMed database. During the search process, English keywords related to the subject, such as *behavioral addiction, neurobiology, reward system, dopamine, glutamate, GABA, serotonin, endocannabinoid system, gambling disorder, gaming disorder, social media addiction, compulsive sexual behavior disorder, food addiction,* and *shopping addiction*, as well as different combinations of these keywords, were used. All figures and graphs presented in this study were originally prepared for the purpose of synthesizing and visualizing the scientific evidence based on the data obtained within the scope of the literature search.

In the literature selection, preference was given, as far as possible, to current studies published within the last five years; however, some older studies that explain the fundamental neurobiological mechanisms of the subject and serve as reference works in the field were also included in the study to enable a comprehensive evaluation. In the selection of articles, research studies and review articles that were available in full text, published in peer-reviewed journals, and provided information on the neurobiology of behavioral addictions, related neurotransmitter systems, the reward circuitry, types of behavioral addictions, and current treatment approaches were taken into consideration. The findings obtained were classified and synthesized according to subject headings, and the neurobiological mechanisms of behavioral addictions were evaluated from a holistic perspective in accordance with the current literature.

3. Common Neurobiology of Behavioral Addictions

3.1. Brain Reward System: The Foundations of Behavioral Addiction

Behavioral addictions are neuropsychiatric disorders characterized by the repetitive and uncontrollable continuation of specific behaviors despite their negative consequences. Evidence obtained in recent years has demonstrated that these disorders cannot be explained solely by psychological processes; rather, they are closely associated with structural and functional changes occurring within the brain's reward circuitry. In particular, dysfunction within the reward circuitry is considered one of the fundamental neurobiological mechanisms underlying the development and maintenance of behavioral addictions. [2]

The reward circuitry of the central nervous system is an integrated neural network responsible for the regulation of motivation, reward anticipation, reinforcement learning, and goal-directed behaviors. Current evidence indicates that this network primarily consists of the ventral tegmental area (VTA), the nucleus accumbens (NAc), which is located within the ventral striatum, and the prefrontal cortex (PFC). These regions remain in continuous communication through various neurotransmitter systems, thereby regulating the evaluation of rewards, behavioral selection, and reinforcement. Neurobiological changes occurring within this circuitry in individuals with behavioral addictions have been reported to contribute to the emergence and persistence of addictive behaviors. [2]

Although dopamine was considered for many years to be the primary regulator of the reward circuitry, current research indicates that this approach is insufficient. The neurobiology of behavioral addictions cannot be explained solely by dopaminergic signaling; other neurotransmitters, such as serotonin, glutamate, and endogenous opioid systems, have also been reported to play important roles in the reinforcement of behaviors by interacting with one another within the reward circuitry. This multidimensional neurotransmitter network contributes to a more comprehensive understanding of the mechanisms underlying the development of behavioral addictions and facilitates the identification of novel therapeutic targets. [2]

Functional neuroimaging studies have demonstrated that the reward circuitry is influenced not only by its core structures but also by its connections with structures of the limbic system. In a resting-state functional magnetic resonance imaging study conducted in adolescents with nonsuicidal self-injurious behavior exhibiting behavioral addiction characteristics, significant changes were detected in functional connectivity associated with the hippocampus, nucleus accumbens, and amygdala. In particular, increased connectivity between the hippocampus and the caudate nucleus, accompanied by decreased connectivity between the nucleus accumbens and the medial cingulate gyrus and between the amygdala and the orbitofrontal cortex, suggests that the network organization of the reward circuitry may be associated with behavioral addiction characteristics. Furthermore, the positive association between increased hippocampus-caudate connectivity and the frequency of self-injurious behavior suggests that functional connectivity changes within the reward circuitry may be associated with the severity of the behavior. [1]

The clinical significance of the reward circuitry is not limited to neuroimaging findings. Neuromodulation studies conducted in obsessive-compulsive-related disorders have demonstrated that stimulation of the dorsolateral prefrontal cortex (DLPFC) using repetitive transcranial magnetic stimulation (rTMS) may influence the reward network associated with the nucleus accumbens and ventral tegmental area. The fact that this intervention produces a significant reduction in symptom severity supports the possibility that the reward circuitry may share common neurobiological mechanisms with behavioral addictions and that this network may be considered a potential therapeutic target. [3]

When the existing findings are considered collectively, it is understood that behavioral addictions are associated not with dysfunction in a single brain region, but rather with alterations in the functional integrity of the cortical and subcortical structures constituting the reward circuitry. The reward network centered on the VTA, nucleus accumbens, and prefrontal cortex operates through interactions with limbic structures and multiple neurotransmitter systems to regulate reward processing, motivation, and behavioral control. Disruptions within this network are thought to constitute a common neurobiological basis across different types of behavioral addictions. [1–3] Importantly, these interconnected neural systems do not function independently, but rather form a dynamic network in which alterations in one component may influence the activity of other regions. The balance between reward-related signaling and higher-order cognitive control is particularly relevant to the persistence of maladaptive behaviors. Accordingly, behavioral addictions may be viewed as disorders involving disrupted communication between reward, motivation, and behavioral control systems. This network-based perspective provides an important framework for understanding the common neurobiological characteristics observed across different behavioral addictions.

The principal neuroanatomical structures involved in behavioral addictions and the dopaminergic connections between these structures are schematically illustrated in Figure 1.

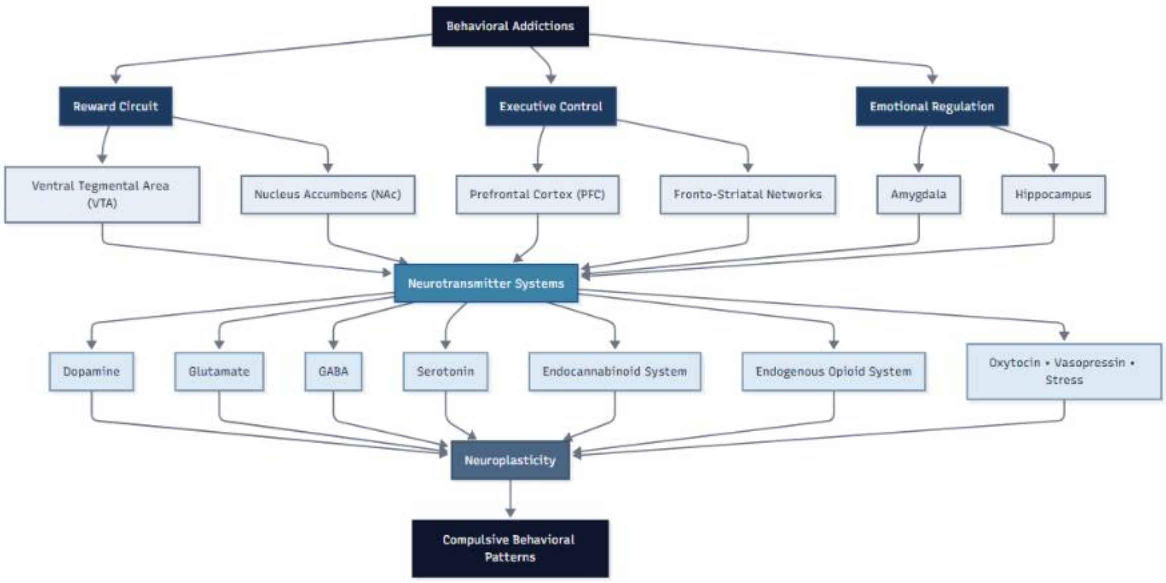


Figure 1. Schematic representation of the mesolimbic reward circuitry involved in the development of behavioral addictions and the associated neurobiological mechanisms.

3.2. Dopamine

For many years, dopamine was regarded as the brain's "pleasure neurotransmitter" and was considered a fundamental biological determinant of addictive behaviors. However, experimental and clinical evidence obtained over the past three decades has demonstrated that the primary function of dopamine is not to generate pleasure, but rather to regulate motivation toward rewards and the incentive salience attributed to reward-related cues. This paradigm shift has enabled the neurobiology of addiction to focus not only on reward consumption but also on the motivational processes that drive reward-seeking behavior. [4]

The Incentive Sensitization Theory (IST) proposes that mesolimbic dopamine systems regulate "wanting" to obtain a reward rather than the hedonic effects of the reward. According to this model, "wanting" and "liking" are separable processes. In addiction, sensitization that develops in the mesolimbic dopamine pathways as a result of repeated exposure causes the individual to develop excessive motivation toward specific rewards or reward-associated cues, whereas the pleasurable effects of the same reward ("liking") may not increase to the same extent. Therefore, as addiction progresses, the individual may continue to seek the reward even if it is no longer as pleasurable as it once was. [4]

This approach is also important in terms of behavioral addictions. Current evidence demonstrates that excessive craving is not specific to substance use disorders but also occurs similarly in gambling and other behavioral addictions. In behavioral addictions, the object of craving is not a substance but the performance of a particular behavior. Therefore, craving is considered not merely a desire to obtain a reward, but rather a powerful motivational state directed toward the completion of a specific action. Recent Bayesian inference-based models propose that this process is continuously updated by comparing the individual's prior expectations with their current internal physiological state, thereby contributing to the persistence of the behavior. [5]

In behavioral addictions, the dopaminergic system is associated with changes not only at the functional level but also at the structural and genetic levels. Meta-analyses examining cortical thickness have demonstrated thinning in the orbitofrontal cortex, dorsolateral prefrontal cortex, precuneus, and postcentral gyrus. In particular, the strong association between dopamine D2 receptor expression and these cortical alterations suggests that dopaminergic signaling plays an important role not only in reward processing but also in cognitive control and behavioral regulation. [6]

On the other hand, the neurobiology of addiction cannot be explained solely in terms of dopamine. Meta-analyses integrating resting-state neuroimaging studies have demonstrated significant alterations in activity in the striatum, anterior cingulate cortex, and ventromedial prefrontal cortex in individuals with behavioral addictions. Molecular analyses, meanwhile, have revealed that, in addition to dopaminergic mechanisms, the serotonergic system is also significantly involved in the process of behavioral addictions. These findings support the view that behavioral addictions are complex neurobiological processes that develop as a result of interactions among multiple neurotransmitter networks. [7]

The temporal characteristics of dopamine signaling are also gaining importance in understanding addictive behaviors. High-temporal-resolution electrochemical recordings conducted in humans have demonstrated that pronounced phasic changes in dopamine occur on subsecond timescales during risky decision-making in individuals with impulse control disorders. In these individuals, dopamine levels increased more prominently during the expectation of positive outcomes, whereas the failure to suppress the dopamine response following outcomes that were worse than expected was noteworthy. These findings demonstrate that dopamine contributes not only to the receipt of rewards but also to the dynamic encoding of reward expectations and the persistence of risky behaviors. [8]

Taken together, the current literature goes beyond the classical view of dopamine as the "pleasure neurotransmitter." While dopamine plays a central role in the development of incentive motivation, strengthening the influence of reward cues on behavior, the development of craving, and the dynamic encoding of reward expectations, the emergence of behavioral addictions is shaped in conjunction with other neurotransmitter networks, particularly the serotonergic system, and prefrontal control mechanisms. [4–8]

3.3. Glutamate

Glutamate is the major excitatory neurotransmitter in the central nervous system and plays a fundamental role in the regulation of learning, memory, synaptic plasticity, and executive functions. Studies conducted in recent years have demonstrated that addiction cannot be explained solely by alterations in the dopaminergic system and that disturbances in glutamatergic transmission also have critical importance in the development and maintenance of addictive behaviors. In particular, the reciprocal interaction between dopamine and glutamate contributes to the emergence of compulsive reward-seeking behaviors by triggering pathological neuroplasticity within the mesolimbic reward system. [9]

One of the most important components of the glutamatergic system in addiction is the corticostriatal pathway. Glutamatergic projections extending from the prefrontal cortex to the nucleus accumbens are involved in the evaluation of environmental cues associated with rewards, the planning of behavior, and the execution of reward-seeking behaviors. Experimental studies have demonstrated that plasticity developing in glutamate transmission within the ventral tegmental area is important for the emergence of addiction-related behaviors, whereas glutamatergic plasticity at the level of the nucleus accumbens plays a determining role in the maintenance and reinstatement of these behaviors. Therefore, dysregulation of corticostriatal glutamate transmission is considered one of the fundamental neurobiological mechanisms underlying addiction. [11]

The effects of glutamatergic signaling on addiction largely arise from its functional interaction with the dopaminergic system. Current reviews report that glutamate and dopamine mutually modulate one another within the mesolimbic system and that this interaction strengthens pathological neuroplasticity. Alterations in glutamatergic activity may affect dopamine release and neuronal responses within the reward circuitry; conversely, changes in dopaminergic activity may also reshape glutamate transmission. This bidirectional relationship is considered one of the fundamental mechanisms explaining the progressive development of increasingly compulsive addictive behaviors over time. [9,12]

During the addiction process, not only glutamate release but also the maintenance of glutamate homeostasis is of critical importance. Under physiological conditions, glutamate levels are precisely regulated by synaptic release, reuptake mechanisms, and glutamate transporters. In particular, impairment of GLT-1-mediated glutamate reuptake has been reported to result in a loss of glutamatergic balance in the nucleus accumbens and ventral tegmental area, which may affect dopaminergic signaling within the reward circuitry and thereby facilitate addiction-related behaviors. Therefore, restoration of glutamate homeostasis is considered one of the potential therapeutic strategies for the prevention of relapse. [12]

When considered in terms of behavioral addictions, the role of the glutamatergic system is particularly supported by studies conducted on gambling disorder. Reviews evaluating pharmacological approaches targeting the glutamate system in gambling disorder have reported that N-acetylcysteine, memantine, amantadine, topiramate, and other glutamate modulators may have beneficial effects on gambling behavior, craving, withdrawal symptoms, and cognitive symptoms. Although the available evidence indicates the need for further controlled studies, regulation of glutamatergic neurotransmission is considered a promising approach in the treatment of behavioral addictions. [10]

The primary reason why the glutamatergic system has gained importance as a therapeutic target is that not only symptoms but also the underlying neurobiological mechanisms of addiction can be targeted. Current literature suggests that agents affecting the glutamatergic system, such as memantine and galantamine, may reduce addictive behaviors by modulating the pathological interaction between the dopaminergic and glutamatergic systems. In particular, regulation of the synergy between glutamate and dopamine may contribute to the development of novel treatment strategies aimed at reducing abnormal neuroplasticity within the reward circuitry. [9]

Overall, glutamate is not merely a secondary neurotransmitter that supports dopamine in the neurobiology of addiction. Rather, it is one of the fundamental neurochemical systems that plays a central role in corticostriatal communication, synaptic plasticity, remodeling of the reward circuitry, and the regulation of relapse mechanisms. Current findings indicate that considering the dopaminergic and glutamatergic systems together will contribute both to a better understanding of the pathophysiology of behavioral addictions and to the development of novel targeted treatment approaches. [9–12]

3.4. GABAergic Regulation in Addiction

GABA (γ-aminobutyric acid) is the principal inhibitory neurotransmitter in the central nervous system and plays a critical role in maintaining the balance of neuronal excitability. Although the dopamine and glutamate systems have frequently been emphasized in the neurobiology of addiction, recent years have shown that GABAergic transmission is an important modulator in the regulation of the reward circuitry and the development of addictive behaviors. In particular, GABAergic neurons in the mesolimbic pathway contribute to the regulation of reward processing, craving, and relapse processes by controlling dopaminergic activity. [13]

Within the mesolimbic reward circuitry, the ventral tegmental area (VTA) constitutes the primary source of dopaminergic neurons, while the activity of these neurons is largely regulated by GABAergic inhibitory inputs. Addictive stimuli may disrupt the physiological balance among GABA, dopamine, and glutamate, thereby altering synaptic plasticity and causing persistent structural and functional changes within the reward circuitry. These changes have been reported to affect not only reward sensitivity but also the persistence of compulsive behaviors and the tendency toward relapse. [13]

GABAB receptors are among the GABA receptor subtypes most extensively investigated in addiction. These metabotropic receptors are involved in the regulation of synaptic inhibition and the maintenance of synaptic plasticity, while also being associated with mood and memory processes. It has been proposed that disruptions in GABAB receptor signaling may constitute common pathophysiological mechanisms in neuropsychiatric disorders such as addiction and depression, and that these receptors may be of importance as therapeutic targets. [14]

Experimental and preclinical studies have demonstrated that ligands and positive allosteric modulators targeting GABAB receptors can reduce the rewarding effects of addictive substances, suppress substance-seeking behavior, and, in some cases, alleviate withdrawal symptoms. These effects are thought to occur largely through inhibitory regulation of dopaminergic neurons in the VTA. Current findings indicate that the GABAergic system is not merely an inhibitory network but also represents one of the potential pharmacological targets in the treatment of addiction. [15]

3.5. Serotonergic Regulation in Addiction

Serotonin (5-HT) is one of the major neurotransmitters involved in mood, impulse control, decision-making, and reward processing. Current studies have demonstrated that the serotonergic system not only modulates dopaminergic activity in addiction but also exerts important effects on craving, relapse, and behavioral control. Therefore, the serotonin system is considered a complementary regulatory mechanism in the neurobiology of addiction. [16,17]

There is a close interaction between the serotonin and dopamine systems. In particular, 5-HT1A and 5-HT1B receptors have been shown to affect dopamine-mediated gene expression in different directions. Experimental findings demonstrate that selective serotonin reuptake inhibitors (SSRIs), when used in combination with methylphenidate, may enhance addiction-related gene regulation and increase cocaine-seeking behavior and the tendency toward relapse. This suggests that alterations in the serotonergic system may influence the risk of addiction. [16]

Different serotonin receptor subtypes perform distinct functions during the addiction process. In particular, studies on nicotine addiction have demonstrated that the 5-HT2C receptor plays an important role in the regulation of nicotine-seeking and relapse behaviors. In contrast, the clinical efficacy of SSRIs has been found to be limited, whereas selective approaches targeting specific serotonin receptors have been reported to yield more promising results. [17]

In recent years, the interaction of the serotonergic system with hormonal regulators has also attracted attention. It has been proposed that sex steroids such as estradiol and testosterone may affect serotonin transporter (SERT) and 5-HT2A receptor expression, which may contribute to sex differences in the development of addiction and treatment response. These findings indicate that the serotonergic system is regulated not only at the neurotransmitter level but also through hormonal mechanisms and may represent an important target for personalized addiction treatments in the future. [18]

3.6. Role of the Endocannabinoid System

The endocannabinoid system (ECS) is one of the neuromodulatory systems that plays an important role in the regulation of reward processing, motivation, impulse control, and synaptic plasticity in the neurobiology of addiction. Consisting primarily of CB1 and CB2 receptors and endogenous ligands such as anandamide and 2-arachidonoylglycerol (2-AG), this system indirectly regulates dopamine, glutamate, and GABA signaling, particularly within the mesolimbic reward circuitry. Therefore, dysfunctions in the ECS contribute not only to substance addictions but also to behavioral addictions such as food addiction by increasing reward sensitivity and facilitating the development of compulsive behaviors. [19]

The CB1 receptor, one of the most important components of the endocannabinoid system, is highly expressed in brain regions critical to addiction, such as the hippocampus, basal ganglia, cerebellum, and prefrontal cortex. CB1 receptors regulate decision-making, impulsivity, and reward anticipation by controlling neurotransmitter release at the presynaptic level. In particular, developmental changes in CB1 receptors during adolescence, together with the incomplete maturation of inhibitory control mechanisms, may increase susceptibility to substance use and the development of addiction. This indicates that the endocannabinoid system is an important determinant not only of reward mechanisms but also of the development of addiction vulnerability. [20]

Endocannabinoid signaling also exerts powerful regulatory effects on synaptic plasticity. Current studies have demonstrated that endocannabinoids modulate synaptic transmission not only between neurons but also through astrocytes. Endocannabinoid signals detected by astrocytes may trigger gliotransmitter release, thereby affecting learning, memory, and experience-dependent synaptic remodeling. These mechanisms are thought to play a role in the development of persistent neuroplastic changes observed during addiction. [21]

3.7. Endogenous Opioid System and Addiction

The endogenous opioid system is an important neuromodulatory network consisting of endorphins, enkephalins, dynorphins, and the μ (mu), δ (delta), and κ (kappa) opioid receptors through which they exert their effects. This system plays a critical role in the regulation of reward perception, the experience of pleasure, motivation, and the stress response. Current studies have demonstrated that alterations in opioid signaling during the addiction process are involved not only in substance use disorders but also in the maintenance of reward-based behavioral addictions. In particular, activation of opioid receptors may increase the hedonic value of rewarding stimuli, thereby contributing to the persistence of compulsive behaviors. [23]

The interaction between the endogenous opioid system and the prefrontal cortex is an important component of the neurobiology of addiction. Opioid peptides and receptors located in the prefrontal cortex influence decision-making, impulse control, motivation, and coping with stress by regulating neuronal excitability, synaptic plasticity, and circuit organization. Dysregulation within this system is thought to contribute to impaired control of reward-oriented behaviors and the development of addiction. In addition, novel therapeutic approaches targeting opioid signaling are considered promising options for addiction and other psychiatric disorders. [22]

3.8. Oxytocin, Vasopressin, and Stress Systems

The neurobiology of behavioral addictions is not limited to classical neurotransmitter systems; oxytocin, vasopressin, and stress-related neuropeptides also play important roles in the regulation of reward processing and addictive behaviors. Oxytocin and vasopressin interact with the mesolimbic dopaminergic system, particularly by modulating social reward processes. Experimental studies have demonstrated that oxytocin may reduce substance use and its rewarding effects, while also supporting the redirection of reward toward social stimuli. Therefore, the oxytocin system is considered one of the promising novel neurobiological targets in addiction treatment. [24]

In contrast, excessive activation of stress systems is considered an important mechanism in the maintenance of addiction. In particular, interactions between corticotropin-releasing factor (CRF) and GABAergic transmission in the central amygdala have been shown to change during alcohol dependence, with CRF sensitivity increasing as addiction develops. These findings support the view that disruption of the balance between reward and stress circuits plays a critical role in the persistence of addictive behaviors. [25] The neurobiology of behavioral addictions arises not from isolated, independent systems but rather from the reciprocal interactions between the reward circuitry and multiple neurotransmitter networks (Figure 2).

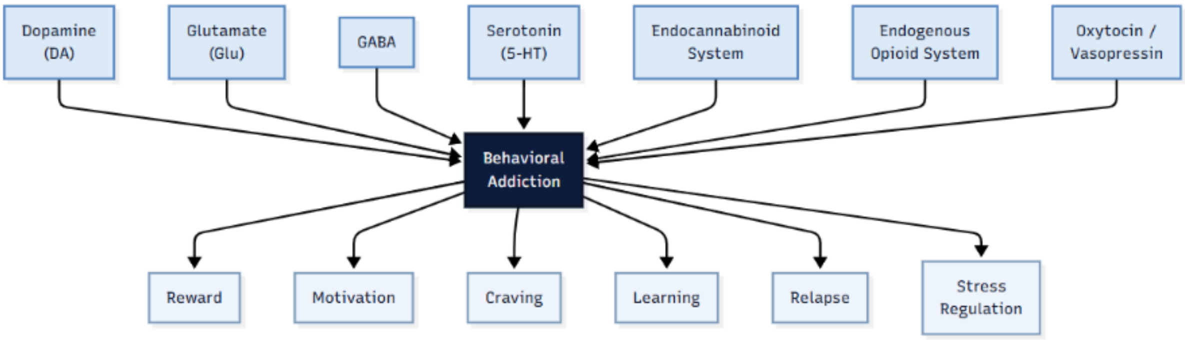


Figure 2. Schematic representation of the relationships among the major neurotransmitter systems, reward circuitry, and behavioral control networks involved in the neurobiological basis of behavioral addictions.

3.9. Fronto-striatal Circuits and Neuromodulation

Fronto-striatal circuits have been shown to play a central role in the development and maintenance of behavioral addictions. Current neuroimaging meta-analyses have demonstrated abnormal activation, particularly in the inferior frontal gyrus, middle frontal gyrus, and caudate nucleus, in individuals with behavioral addictions. These regions constitute key networks that regulate reward seeking, self-control, and decision-making processes, and alterations in their functional connectivity contribute to the development of compulsive behaviors. The findings also support the view that the neurobiology of behavioral addictions is associated with dysfunctions in fronto-striatal networks, consistent with the I-PACE model. [26]

The critical role played by these circuits in addiction has also increased the importance of neuromodulation approaches. In particular, high-frequency repetitive transcranial magnetic stimulation (rTMS) targeting the dorsolateral prefrontal cortex has been reported to have the potential to reduce symptoms of behavioral addictions by regulating reward processing and inhibitory control mechanisms. Current studies conducted in adolescents indicate that rTMS may represent a promising therapeutic approach capable of improving neurophysiological markers and clinical symptoms. [27]

4. Behavioral Addictions

4.1. Gambling Addiction

Gambling addiction is a chronic disorder defined in the DSM-5 as a substance-free behavioral addiction and characterized by the individual's loss of control over gambling behavior. With the widespread availability of online betting platforms and mobile applications, easier access to gambling has increased the prevalence of the disorder. Gambling addiction not only causes financial losses but is also associated with serious psychosocial consequences such as depression, anxiety, suicidal tendencies, family problems, and criminal behaviors. Current evidence indicates that gambling addiction is an important public health problem and that further independent research is needed regarding the epidemiology, neurobiology, and treatment of the disorder. [28]

From a neurobiological perspective, gambling addiction is associated with functional alterations in serotonergic and dopaminergic circuits, particularly those responsible for the reward system and impulse control. Imaging studies have demonstrated increased serotonin transporter activity in the ventromedial prefrontal cortex in individuals with gambling addiction, and this alteration has been found to be positively associated with the level of impulsivity. In addition, striatal dopamine transporter activity has also been associated with impulsivity, and alterations in the serotonergic system have been suggested to play a more prominent role in gambling addiction. These findings suggest that impulsivity may represent not only a clinical symptom but also an important biomarker reflecting the neurobiological basis of addiction. [30]

There is currently no specific FDA-approved pharmacological agent for the treatment of gambling addiction. Therefore, neuromodulation methods have attracted increasing attention in recent years. Continuous theta burst stimulation (cTBS) applied to the pre-supplementary motor area (pre-SMA) significantly reduced gambling craving and symptom severity in a randomized, double-blind, controlled study, while also producing improvements in gambling-related functional impairment. These findings indicate that noninvasive brain stimulation methods targeting prefrontal control networks may represent promising options for the treatment of gambling addiction. [29]

Therefore, gambling addiction is a complex behavioral addiction that develops as a result of the interaction among neurobiological alterations in the reward system, impulsivity, and executive functions. In the future, the identification of neurobiological markers and the development of targeted neuromodulation approaches represent important areas of research that may improve treatment outcomes. [28–30]

4.2. Gaming Addiction

Gaming addiction is defined as a behavioral addiction in the ICD-11 and is associated with neurobiological alterations occurring in brain regions responsible for reward processing, impulse control, and executive functions. In particular, functional alterations observed in the dorsolateral prefrontal cortex, anterior cingulate cortex, striatum, orbitofrontal cortex, insula, and amygdala play important roles in loss of control, excessive gaming behavior, and the continuation of gaming despite negative consequences. Nevertheless, debates continue regarding the classification of gaming addiction as a psychiatric disorder. [31]

A strong association between ADHD and gaming addiction has been reported in children and adolescents. Dysregulation of the reward system, impulsivity, and impairments in attentional control observed in ADHD increase the risk of developing gaming addiction. Therefore, ADHD is considered an important risk factor for gaming addiction, particularly in younger age groups, and it is recommended that it be evaluated concurrently during the diagnostic and treatment process. [32]

The strongest evidence for the treatment of gaming addiction is available for cognitive behavioral therapy (CBT). In addition, mindfulness-based approaches, family therapy, and relapse prevention programs have demonstrated promising results. In cases of comorbid conditions such as depression, pharmacological treatments such as bupropion and escitalopram may be utilized, while novel treatment approaches targeting brain activity through neuromodulation methods such as rTMS and tDCS are also being investigated. Functional imaging studies indicate that these treatments may contribute to normalization within reward and executive control networks. [33]

4.3. Social Media Addiction

Social media addiction is one of the behavioral addictions characterized by an individual's uncontrollable use of social networking platforms, inability to reduce the duration of use, and the negative impact of this behavior on academic, occupational, or social functioning. Studies conducted in recent years have demonstrated that problematic social media use is not merely a behavioral habit but rather a neurobiological process associated with alterations in reward systems, impulse control, and emotion regulation mechanisms. In particular, repeated stimulation of dopaminergic reward circuits reinforces social media use, while weakened inhibitory control in the prefrontal cortex contributes to the persistence of addictive behavior. In addition, problematic social media use has been reported to be associated with depression, anxiety, stress, obsessive-compulsive symptoms, attention-deficit/hyperactivity disorder, FOMO (fear of missing out), and sleep disorders. Neuroimaging studies have also revealed structural and functional alterations in reward- and emotion-regulation regions such as the nucleus accumbens, amygdala, and insula. [35]

Reliable assessment of social media addiction is important for both clinical practice and epidemiological research. For this purpose, a study examining the Swedish adaptation of the Bergen Social Media Addiction Scale (BSMAS) for adolescents found that the scale demonstrated high reliability and validity. The study showed that core addiction symptoms were significantly associated with anxiety and depression symptoms and that the scale demonstrated measurement invariance with respect to sex and country of birth. These findings support the BSMAS as a reliable instrument for the early screening of social media addiction and the identification of at-risk individuals among adolescents. [34]

4.4. Compulsive Sexual Behavior Disorder

Compulsive Sexual Behavior Disorder (CSBD) is defined in the ICD-11 as a behavioral addiction that causes marked impairment in personal, social, and occupational functioning as a result of the individual's inability to control intense and repetitive sexual urges, thoughts, and behaviors. Studies conducted in recent years have demonstrated that CSBD is not limited to behavioral symptoms but also involves mechanisms observed in addictive disorders, such as tolerance, withdrawal symptoms, and alterations in the reward system. Particularly when evaluated in conjunction with problematic pornography use, an increasingly greater need for sexual stimuli, difficult-to-control sexual thoughts, irritability, mood changes, and sleep problems have been reported to be significantly associated with disease severity. [36]

The clinical presentation of CSBD is not uniform, and different symptom dimensions may affect an individual's sexual functioning in different ways. Research has demonstrated that dimensions such as salience and relapse are associated with increased sexual desire and arousal, whereas dimensions involving negative consequences and dissatisfaction are associated with sexual dysfunction, difficulty achieving orgasm, and reduced sexual satisfaction. Therefore, it is emphasized that the assessment of CSBD should examine not only overall symptom severity but also each symptom cluster separately. [37]

Neuroimaging studies also provide important findings supporting the neurobiological basis of CSBD. Structural and functional magnetic resonance imaging studies have identified differences in structural volume and functional connectivity in brain regions associated with reward processing, impulse control, and emotional regulation, such as the cerebellum, orbitofrontal cortex, amygdala, and prefrontal cortex. In particular, reductions in connectivity involving the orbitofrontal cortex and increases in amygdala connectivity have been suggested to be associated with increased sensitivity to sexual stimuli and impairments in impulse control. These findings support the view that CSBD is not only psychological in nature but also has distinct neurobiological underpinnings. [38]

Current literature indicates that biological and environmental factors jointly contribute to the development of CSBD. Although a higher prevalence has been reported among men, neuroticism, stress sensitivity, and childhood trauma are reported to contribute more substantially to the development of symptoms in women. In addition, impulsivity and psychiatric comorbidities are commonly observed in both sexes. [39] Furthermore, the presence of compulsive sexual behaviors among the impulse control disorders that develop in patients with Parkinson's disease as a result of dopaminergic treatments suggests that dysregulation of the dopaminergic reward system plays an important role in the emergence of this behavior. These observations in patients with Parkinson's disease contribute to the understanding of the neurobiological mechanisms of CSBD and may provide insight into diagnostic and therapeutic approaches to be developed in the future. [40]

4.5. Neurobiology of Romantic Love and Attachment

Romantic love is not merely an emotional experience but is also closely associated with neurobiological processes that activate reward and motivational systems. In particular, activation of dopaminergic reward circuits may cause romantic love to exhibit addiction-like characteristics.

Therefore, some researchers have proposed that common neurobiological mechanisms exist between intense romantic attachment and substance addiction. In both conditions, reward system activation, dopamine release, and motivational processes play important roles. [41]

Functional neuroimaging studies have demonstrated that a romantic partner is represented preferentially within the brain's reward system. In particular, neural patterns associated with a romantic partner have been shown to be distinguishable from those associated with unfamiliar individuals in the nucleus accumbens, supporting the strong relationship between romantic attachment and reward circuits. These findings demonstrate that dopaminergic reward mechanisms play a fundamental role in maintaining romantic relationships. [42]

Oxytocin also emerges as an important neurobiological regulator in the maintenance of romantic attachment. In individuals receiving intranasal oxytocin, reduced interest toward strangers and other individuals in their close social environment has been demonstrated, whereas behaviors supporting the exclusivity of the romantic relationship have increased. In addition, this effect has been reported to be more pronounced particularly during the early stages of the relationship and may be influenced by the individual's life history and childhood experiences. [43]

Taken together, the available findings indicate that the phenomenon defined as romantic love and attachment is associated not only with psychological processes but also with neurobiological mechanisms such as the reward system, dopaminergic transmission, and oxytocinergic regulation. Nevertheless, although current evidence demonstrates similarities between the physiological characteristics of romantic love and pathological addiction processes, further clinical and neurobiological research is needed to better understand the diagnostic boundaries and underlying mechanisms of romantic love and attachment. [41–43]

4.6. Food Addiction

Food addiction is a controversial concept characterized by individuals consuming particularly high-calorie and highly palatable foods beyond their energy requirements while experiencing a loss of control. Although food addiction has not been defined as an independent disorder in the DSM-5 and DSM-5-TR, studies have demonstrated that certain foods may affect the reward system in a manner similar to substances of abuse. Within this context, the Yale Food Addiction Scale (YFAS) has been developed and is used to assess addiction-like behaviors associated with food by utilizing criteria derived from substance use disorders. In particular, alterations in dopaminergic reward pathways are thought to play an important role in the development of craving for highly palatable foods and loss of control. [44]

The neurobiological basis of food addiction involves brain regions associated with reward, motivation, and the regulation of feeding behavior. Regions such as the ventral tegmental area (VTA), amygdala, and orbitofrontal cortex (OFC) are involved in the evaluation of food-related cues and the formation of reward anticipation. In particular, VTA-derived dopaminergic signals have been shown to increase the salience of food-related stimuli, thereby potentially strengthening addiction-like eating behaviors. In addition, disruption of the balance between homeostatic feeding mechanisms regulated by the hypothalamus and the reward system may contribute to the development of overeating behaviors. [44,45]

Genetic susceptibility, environmental factors, and persistent consumption of highly palatable foods all contribute to the processes of overeating and food addiction. Prolonged exposure to rewarding foods may lead to alterations in synaptic plasticity, thereby modifying the functioning of the reward system and reinforcing consumption behavior. Currently, novel therapeutic approaches such as GLP-1 receptor agonists are being investigated because of their effects on both metabolic regulation and appetite and reward mechanisms. However, further studies are needed to classify food addiction as a clinical disorder and to determine effective treatment methods. [45]

4.7. Shopping Addiction

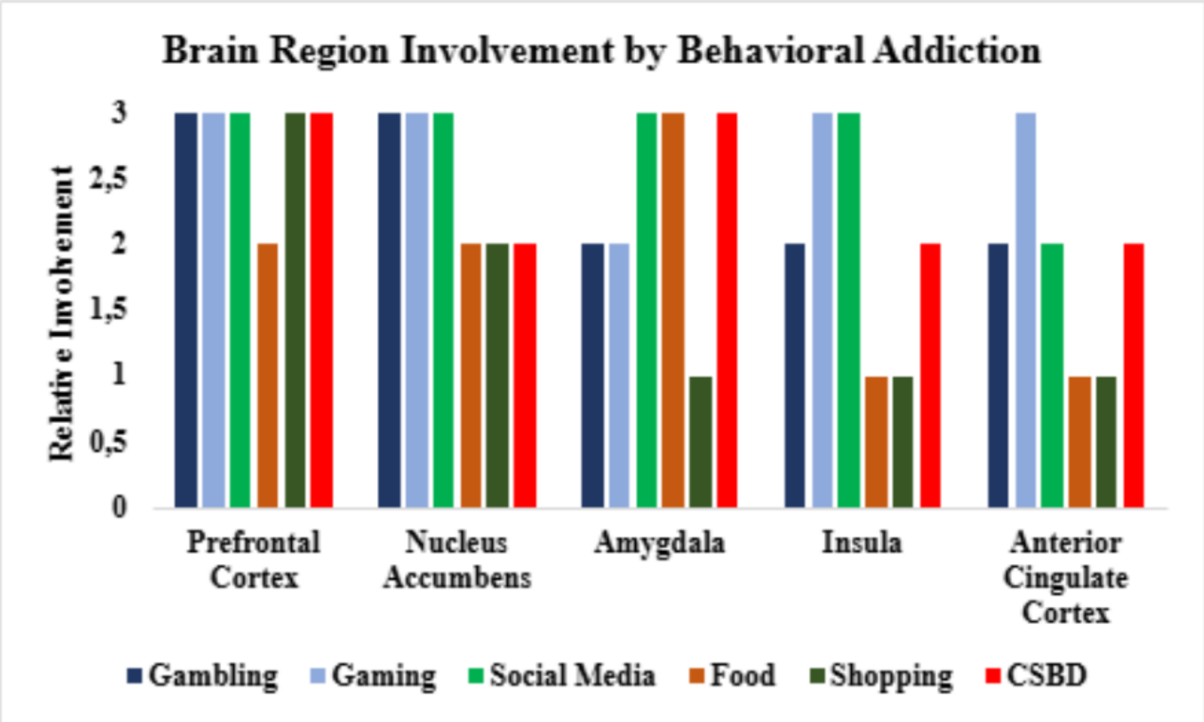
Shopping addiction (compulsive buying disorder) is a type of behavioral addiction characterized by shopping behaviors that exceed the amount required by the individual and are difficult to control, potentially leading to significant psychological distress and functional impairment. Although this condition was previously considered within the context of impulse control disorders or the obsessive-compulsive spectrum, it is currently addressed within the framework of behavioral addictions. Shopping addiction is frequently associated with psychiatric conditions such as depression, anxiety disorders, substance use disorders, and personality disorders. Its prevalence in the general population has been reported to be approximately 5%, with onset most commonly occurring during young adulthood. However, the neurobiological basis of the disorder and standardized treatment approaches remain insufficiently elucidated. [46]

Psychological factors play an important role in the development of shopping addiction. This behavior is often performed with the aim of experiencing pleasure, avoiding negative emotions, or coping with stress. However, although shopping may produce short-term relief and a sense of reward, feelings of guilt, regret, and loss of control may subsequently emerge. Studies have demonstrated that shopping addiction is particularly associated with impulsivity and certain personality traits. Difficulties in impulse control and characteristics such as extraversion may contribute to shopping behavior becoming repetitive and difficult to control. [47]

From a neurobiological perspective, shopping addiction, similar to other behavioral addictions, is associated with dysfunctions in brain systems involved in reward processing, decision-making, and behavioral control. Systematic reviews have demonstrated impairments in executive functions, increased reward sensitivity, and an increased tendency to respond to shopping-related cues in compulsive buying behavior. In particular, disruption of the functional balance between the striatum and frontal cortex may be associated with reduced impulse control and reinforcement of rewarding shopping behaviors. These findings demonstrate that shopping addiction is a complex behavioral addiction with not only psychological but also neurobiological underpinnings. [48]

5. Comparative Neurobiology

Although behavioral addictions occur without substance use, they share many neurobiological characteristics with substance use disorders. Current evidence indicates that disorders such as gambling, gaming, compulsive buying, problematic social media use, and compulsive sexual behavior affect common neural networks responsible for reward processing, impulse control, and decision-making. In particular, the mesolimbic dopaminergic system, ventral striatum, prefrontal cortex, and limbic structures play central roles in reward anticipation, craving, and the persistence of behavior in both substance and behavioral addictions. However, the level of diagnostic evidence differs among behavioral addictions; while gambling disorder is recognized as an addiction in the DSM-5 and ICD-11, further neurobiological and clinical research is needed for conditions such as compulsive buying disorder and problematic social media use. [49] The distribution and relative weighting of affected brain regions across different types of behavioral addictions are comparatively presented in Graph 1.



Graph 1. Comparison of the relative levels of involvement of major brain regions across different types of behavioral addictions. The values represent a qualitative synthesis of neuroimaging and neurobiological findings reported in the current literature.

The common neurobiological characteristics of behavioral addictions include increased reward sensitivity, impairments in executive functions, increased impulsivity, and reduced cognitive control over behavior. These disorders frequently co-occur with psychiatric conditions such as depression, anxiety disorders, and attention-deficit/hyperactivity disorder and, when left undiagnosed, can lead to serious adverse consequences for functioning. Studies conducted in psychosomatic patient groups have also shown that symptoms of behavioral addictions are more prevalent than expected, with shopping addiction being particularly frequent, indicating that behavioral addictions should also be screened for during routine psychiatric evaluations. [49,51]

The emergence of behavioral addictions cannot be explained solely by neurobiological mechanisms; digitalization, socioeconomic conditions, and cultural factors also play important roles. In particular, with the widespread availability of internet access and online platforms, the prevalence of addictive behaviors related to gaming, gambling, and social media use has increased. However, the standardization of diagnostic tools, insufficient epidemiological data, and limited availability of treatment services remain major challenges in many countries. Currently, there is no approved specific pharmacological treatment for behavioral addictions, and the strongest evidence supports cognitive behavioral therapy. In the future, a better understanding of the underlying neurobiological mechanisms and an increase in research taking cultural differences into account will contribute to the development of both diagnostic and treatment approaches. [49–51]

6. New Perspectives

Current research on the neurobiology of addiction is moving beyond classical approaches focused exclusively on the dopaminergic reward system and is bringing new mechanisms such as metabolic regulation, neuroinflammation, gut microbiota, and artificial intelligence-supported digital environments to the forefront. In particular, glucagon-like peptide-1 (GLP-1) receptor agonists (GLP-1RAs) have been shown to modulate dopaminergic, glutamatergic, and GABAergic neurotransmission through receptors located in the ventral tegmental area (VTA), nucleus accumbens (NAc), and prefrontal cortex (PFC), thereby reducing drug-seeking behavior and craving. In addition, the ability of vagal signaling, hormonal regulation, and neuroinflammatory processes through the gut–brain axis to influence reward mechanisms suggests that addiction should be considered not only a disorder of the central nervous system but also a systemic biological disorder. Preclinical and early-stage clinical studies have demonstrated that GLP-1 receptor agonists may reduce the desire to use substances such as alcohol, nicotine, and cocaine, providing promising findings for future treatment strategies. [52]

Another development that has attracted attention in recent years is the influence of neuroinflammation and the gut microbiota on addiction. Experimental studies have shown that morphine exposure increases Drp1-mediated mitochondrial fragmentation in astrocytes in the nucleus accumbens, thereby enhancing neuroinflammation and addictive behaviors. Inhibition or gene silencing of Drp1, in contrast, significantly reduced both the inflammatory response and drug-seeking behavior. In addition, alterations in the gut microbiota may influence the development of addiction through the immune system, metabolic pathways, and nervous system; probiotic administration, fecal microbiota transplantation, and microbiome-based therapies have therefore been proposed as potential adjunctive approaches in the future. The relationship of the microbiome with stress, anxiety, depression, and other psychiatric comorbidities further increases the importance of personalized approaches in addiction treatment. [53–55]

Technological developments have also added a new dimension to addiction research. In particular, artificial intelligence-supported social media algorithms analyze user behavior, provide personalized content, and continuously stimulate the reward system, thereby increasing dopaminergic activation. This may lead to changes in the functions of the prefrontal cortex and amygdala, which are responsible for attentional control, decision-making, and emotional regulation, particularly in adolescents, and may increase the risk of behavioral addiction. Today, the neurobiology of addiction is moving toward a multidimensional perspective that considers the effects of synaptic plasticity, neuroinflammation, metabolic signaling, the gut–brain axis, and the digital environment together. These developments hold significant potential for the future development of biomarker-based diagnostic methods and the establishment of personalized pharmacological and behavioral treatment approaches. [52–56] The neurobiological processes discussed in the preceding sections demonstrate that behavioral addictions cannot be explained by a single brain region; rather, they emerge through complex interactions among reward, motivation, cognitive control, and emotion regulation systems. This network of multiple interactions is schematically presented in Graph 2.

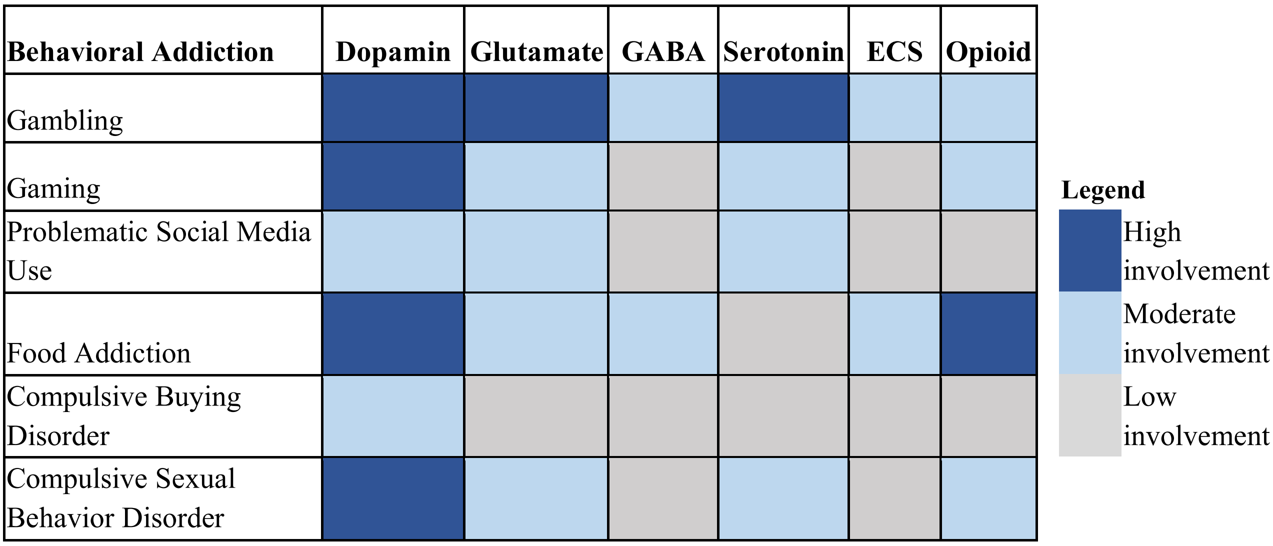
Characteristic	Gambling Disorder	Gaming Disorder	Problematic Social Media Use	Food Addiction	Compulsive Buying Disorder	Compulsive Sexual Behavior Disorder
Reward Circuit Dysfunction	Present	Present	Present	Present	Present	Present
Dopaminergic Alteration	High	High	Moderate	High	Moderate	High
Glutamatergic Involvement	High	Moderate	Moderate	Moderate	Low	Moderate
GABAergic Involvement	Moderate	Low	Low	Moderate	Low	Low
Serotonergic Involvement	High	Moderate	Moderate	Low	Low	Moderate
Endocannabinoid System	Moderate	Low	Low	Moderate	Low	Low
Endogenous Opioid System	Moderate	Moderate	Low	High	Low	Moderate
Oxytocin / Vasopressin	Low	Low	Moderate	Low	Low	Low
Executive Dysfunction	High	High	High	Moderate	High	High
Impaired Decision Making	High	High	Moderate	Moderate	High	High
Craving	High	High	High	High	Moderate	High
Neuroplastic Changes	High	High	Moderate	Moderate	Moderate	Moderate
Neuroimaging Evidence	Strong	Strong	Moderate	Moderate	Limited	Moderate

Graph 2. Comparative summary of the common neurobiological features reported across major types of behavioral addictions based on the recent literature.

7. Discussion

This review comprehensively evaluated the neurobiological foundations of behavioral addictions in light of the current literature. The findings indicate that behavioral addictions cannot be explained solely by psychological or sociocultural factors; rather, they emerge from complex neurobiological processes involving the coordinated dysfunction of multiple neural networks responsible for reward processing, motivation, learning, impulse control, and executive functions. In particular, functional alterations within the reward circuitry centered on the ventral tegmental area, nucleus accumbens, and prefrontal cortex represent common neurobiological features across different types of behavioral addictions. [2,4,26,49]

For many years, the neurobiology of addiction was largely explained by dopamine-based models. However, evidence accumulated in recent years indicates that dopamine represents only one component of the addiction process, while glutamatergic, GABAergic, serotonergic, endocannabinoid, and endogenous opioid systems, together with oxytocin, vasopressin, and stress-related mechanisms, continuously interact with the reward circuitry. [4,9,13,16,19,22,24] Accordingly, addiction is now increasingly conceptualized not as a disorder involving a single neurotransmitter system, but rather as a systems-level disorder characterized by dysfunction across multiple neurotransmitter networks and the functional circuits formed by their interactions. The relative contributions of different neurotransmitter systems may vary among behavioral addiction types, and these relationships are summarized in Graph 3.



Graph 3. Heatmap summarizing the relative involvement of major neurotransmitter systems across different types of behavioral addictions based on the current literature.

Although substantial common neurobiological features were identified across the behavioral addictions examined, important differences were also observed. The evidence derived from neuroimaging, genetic, and neuromodulation studies is relatively robust for gambling disorder and gaming disorder, both of which are recognized as behavioral addictions within international diagnostic classification systems. [28–33] In contrast, a substantial proportion of the existing studies on conditions such as social media addiction, compulsive buying disorder, and romantic love and attachment employ cross-sectional designs, and no consensus has yet been established regarding their diagnostic criteria or reliable biomarkers. [34,35,41,46,49] These findings indicate that although behavioral addictions may share common biological mechanisms, they do not all have the same level of clinical evidence.

Another notable aspect of this review is the progressive expansion of addiction research from the molecular level toward systems biology. Evidence concerning the effects of GLP-1 receptor agonists, the gut–brain axis, neuroinflammation, the microbiota, and artificial intelligence-driven digital environments on the reward system indicates that the neurobiology of addiction is extending beyond classical dopaminergic approaches. [52–56] In particular, a better understanding of the interactions between metabolic signaling, the immune system, neuroinflammatory mechanisms, and the reward circuitry may contribute to the identification of shared biological targets for both behavioral and substance addictions in the future.

Despite the promising findings provided by the current literature, several important limitations should be acknowledged. Most studies have been conducted in relatively small sample populations, and substantial heterogeneity exists in the diagnostic criteria and assessment methods used across different behavioral addictions. Furthermore, because a considerable proportion of the available evidence is derived from cross-sectional designs, it remains difficult to determine whether the observed neurobiological alterations are causes or consequences of addictive behavior. Inadequate control of variables such as age, sex, genetic predisposition, psychiatric comorbidities, and environmental factors further limits the generalizability of the findings. [31,39,49]

Future research integrating multicenter prospective studies, longitudinal neuroimaging follow-up, multi-omics approaches, and artificial intelligence-assisted data analysis may contribute to a more detailed understanding of the neurobiological mechanisms underlying behavioral addictions. In particular, the development of biomarker-based diagnostic methods, identification of distinct behavioral addiction subtypes, and establishment of individualized pharmacological or neuromodulation-based treatment strategies are likely to represent major research priorities in the coming years. [27,52–56] These multilayered interactions can be summarized within an integrated model of the neurobiological mechanisms involved in the development of behavioral addictions (Figure 3).

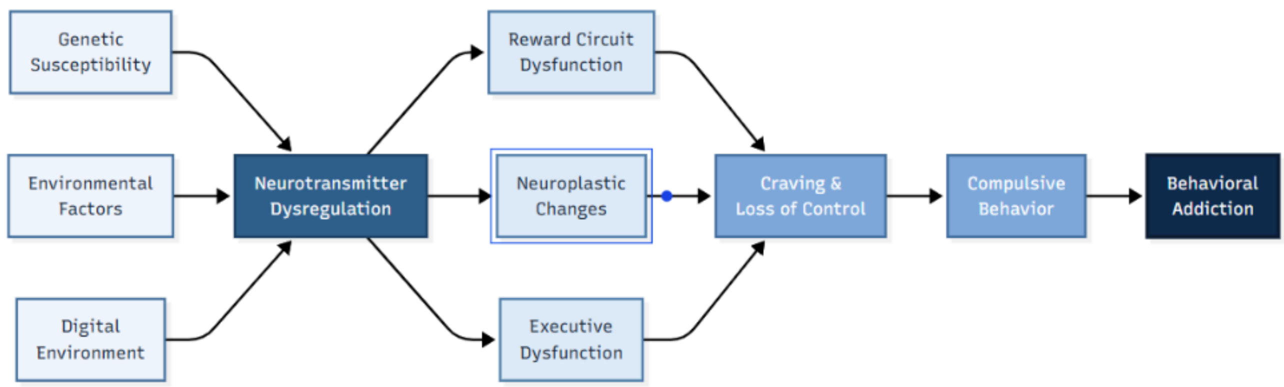


Figure 3. Integrated schematic model of the neurobiological mechanisms involved in the development of behavioral addictions.

8. Conclusion

Behavioral addictions are complex neuropsychiatric disorders shaped by multidimensional neurobiological mechanisms that extend beyond the uncontrolled repetition of specific behaviors.

Current evidence indicates that, in addition to the mesolimbic reward system and frontostriatal networks, dopaminergic, glutamatergic, GABAergic, serotonergic, endocannabinoid, and endogenous opioid systems interactively contribute to the development of these disorders. Regulatory mechanisms involving oxytocin, vasopressin, stress responses, and synaptic plasticity also make important contributions to the emergence and maintenance of behavioral addictions.

Although different types of behavioral addictions, including gambling disorder, gaming disorder, social media addiction, compulsive sexual behavior disorder, romantic love and attachment, food addiction, and compulsive buying disorder, share several common neurobiological features, each also has distinct clinical and biological characteristics. Therefore, behavioral addictions cannot be adequately explained by a single underlying mechanism, and biological, psychological, and environmental factors should be considered together during assessment.

In recent years, research into the effects of neuroinflammation, the gut–brain axis, the microbiota, GLP-1 receptor agonists, and artificial intelligence-driven digital environments on addiction has provided new perspectives on the understanding of behavioral addictions. These developments hold considerable potential for the future development of biomarker-based diagnostic approaches, biological stratification of patients, and implementation of personalized treatment strategies.

In conclusion, elucidating the neurobiological mechanisms underlying behavioral addictions in greater detail may contribute substantially to improving early diagnosis, developing targeted therapeutic strategies, and reducing the burden of these disorders on individuals and society.

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Conflict of Interest Statement

No conflict of interest has been declared.